

Anticipation Effects in Difference-in-Differences Models^{*}

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Abstract

Difference-in-differences (DID) typically relies on two key assumptions: parallel trends and no anticipation. The existing literature focuses mainly on the credibility of parallel trends, while the no-anticipation assumption is often maintained with little empirical scrutiny. This paper studies identification in DID models in the presence of anticipation effects. Under parallel trends alone, we show that the standard DID estimand identifies the average treatment effect on the treated (ATT) net of the average anticipation effect. We derive bounds on the ATT when the average anticipation effect evolves exponentially over time and is bounded in magnitude by the ATT. The bounds take the form of intersection-union bounds, and inference can be implemented using existing methods. We also derive a misspecification-robust bound when it is unknown which identifying assumption fails. In settings with multiple treatment periods, we establish a testable implication of the identifying assumptions and extend the identification analysis accordingly. Finally, we illustrate the usefulness of our bounds through several empirical examples.

Keywords: Difference-in-differences, anticipation effects, partial identification.

JEL classification: C12, C21, C23

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1 Introduction

Difference-in-differences (DID) is one of the most popular research designs for estimating causal effects in non-experimental settings. One of the parameters of interest in this context is known as “the average treatment effect on the treated (ATT)”, i.e., we are interested in evaluating the treatment effect for the units that are in the treatment group. To identify ATT, we typically rely on two assumptions: the “Parallel trends” assumption (PT) and the “No-anticipation effects” assumption (No-AE). The parallel trends assumption means that the average outcome among the treated and untreated populations would have followed the same trends in the absence of treatment. The no-anticipation assumption posits that the outcome before the treatment does not depend on future treatment, in other words, people do not anticipate or react to forthcoming treatments.

Despite having these two identifying assumptions, people are usually concerned about the credibility of the parallel trends assumption. To assess this, they examine multiple pre-treatment periods to determine if the treatment and control groups exhibit comparable trends before treatment. If so, people will say “pre-trend” holds, providing plausible justification or evidence for the parallel trends assumption, though it does not guarantee that the parallel trends assumption is satisfied, given its counterfactual nature and non-testability. If the “pre-trend” fails, people often attribute this to the violation of parallel trends and turn to some identification result which is robust to the violation of parallel trends. However, the failure of the pre-trend can be driven by two sources: (i) the violation of parallel trends and (ii) the violation of no-anticipation effects.

If the no-anticipation assumption (often implicitly assumed) is true, it implies that individuals do not act on the knowledge of their future treatment before treatment starts. Nevertheless, anticipation becomes plausible when individuals are forward-looking and have access to information on future treatment, and there is a benefit to acting before treatment is adopted. For instance, when a policy announcement precedes its implementation, some forward-thinking individuals may strategically adjust their behavior after the announcement date. This could potentially impact outcomes, resulting in a distortion in the estimate of the treatment effect in DID setting. To address this issue, researchers usually either drop some pre-treatment periods that might have been contaminated by anticipation effects or use the announcement date as the treatment date, obtaining a combined effect of the announcement and the implementation of the policy. However, it’s important to note that the timing of anticipation may vary across units, and people may anticipate treatment even before the announcement date. Some individuals may be more informed than others, they may have inside information or see rumors and newspapers discussing the policy, prompting them

to change their behavior even earlier. Hence, to make it clear, we assume throughout the paper: (i) Researchers have no information about when people start to anticipate, and (ii) We are interested in the implementation effect (effectiveness of policy) after the policy is implemented.

Neglecting the anticipation effects may lead to misleading conclusions. We show that the standard DID estimand identifies the net effect of the treatment on the treated group: ATT minus the average anticipation effect (AE). There is also some empirical evidence suggesting that the presence of anticipation can bias the estimation of treatment effects. [Alpert \(2016\)](#) finds that individuals reduced drug utilization for chronic drugs in anticipation of Medicare Part D’s implementation in 2014 (which was aimed at lowering the cost of prescription medications for Medicare beneficiaries). Without accounting for this anticipatory response, the estimated treatment effect of Part D on drug utilization will be substantially overestimated. [Malani and Reif \(2015\)](#) finds that physician labor supply will increase before the tort reform (which was not targeted at physicians but was likely anticipated by them). Researchers without considering anticipation effects might underestimate the effects of reform on labor supply or even potentially misinterpret the direction of the effect. Therefore, it is imperative to allow for anticipatory behaviors in the model. Given the motivation above, the goal of our paper is to relax the no-anticipation effect assumption and establish an identified set for ATT.

In this paper, we ask what can be learned about the parameter of interest if the no-anticipation assumption fails. First, we provide identification results for the parameter of interest, more specifically, we provide a bound on the ATT under the assumption that the average anticipation effect is bounded in magnitude by the true ATT. By incorporating information from multiple pre-treatment periods, we can tighten this bound under the assumption that the average anticipation effect is bounded and exponentially discounted. Here, the way to model the anticipation behavior is motivated by the exponential discounting looking-forward model, which is commonly used in finance and macroeconomics ([Malani and Reif, 2015](#)). Then, estimation and inference for our bounds are provided and discussed. Second, we combine our bound with the generalized DID bound (GDID) developed in [Ban and Kedagni \(2022\)](#), resulting in a misspecification robust bound. Third, we move on to the DID model with multiple treatment periods and develop a testable implication for the extended PT and the extended No-AE. Additionally, we give an identification result under a relaxed assumption in a staggered adoption design. Finally, we illustrate our methods by using datasets from [Kresch \(2020\)](#) and [Ladino, Saavedra, and Wiesner \(2021\)](#).

This paper contributes to the current DID literature by explicitly examining and

incorporating anticipation effects. In terms of relaxing identifying assumptions, most papers are concerned about the parallel trends assumption. [Callaway and Sant’Anna \(2021\)](#) deals with cases where the parallel trends assumption only holds after conditioning on some observed covariates. [Manski and Pepper \(2018\)](#), [Rambachan and Roth \(2019\)](#) and [Ban and Kedagni \(2022\)](#) work on strategies to partially identify the causal effect parameters when the parallel trends assumption does not hold. However, limited attention has been given to the “no-anticipation” assumption, despite its significance. Our paper addresses this gap by relaxing no-anticipation effects and adopting the bounded variation assumption for anticipation effects. We derive bounds on ATT from this weaker assumption and provide a practical method for estimation and inference.

Moreover, our paper extends the analysis to multiple treatment periods, aligning with the recent trend in DID literature which focuses on allowing for multiple post-treatment periods and variation in treatment timing. When there is heterogeneity in treatment effects over time, the interpretation of parameters associated with standard two-way fixed effect (TWFE) specification would be incorrect, as shown in [Goodman-Bacon \(2021\)](#) and [De Chaisemartin and d’Haultfoeuille \(2020\)](#). And then [Callaway and Sant’Anna \(2021\)](#), [Sun and Abraham \(2021\)](#), [De Chaisemartin and d’Haultfoeuille \(2020\)](#), [Borusyak, Jaravel, and Spiess \(2021\)](#) and [Wooldridge \(2021\)](#) propose alternative estimators that bypass the pitfall associated with TWFE. Building on the same setting, we find that the TWFE estimate would be biased if people start to anticipate before the initial treatment of the “first treated group”. We also show that the extended PT and the extended No-AE can be tested jointly in the multiple treatment period setting, and we give the identification result when the extended No-AE is relaxed in the staggered adoption design.

Related Literature This paper is closely related to the work of [Malani and Reif \(2015\)](#) and [Gong \(2021\)](#). [Malani and Reif \(2015\)](#) provides a framework for estimating the anticipation effects. They assume a strong parametric form for the outcomes, where the outcomes depend on expectations of future treatment, and expectations are discounted at an exponential rate. This assumption is motivated by a forward-looking model as well as the convenience of estimation. Our paper only assumes the magnitude of anticipation effects is bounded by exponentially discounted future treatment effects, which is a relaxed version of their assumption. Also, they use a set of instrumental variables to address the problem of unobservable expectations, while our paper does not require it. [Gong \(2021\)](#) provides the bounds on ATT by using a similar assumption as our paper. Gong (2021) divides treated people into two types, anticipators and non-anticipators, and assumes that the magnitude of anticipation effects by anticipators is bounded by the true ATT, and the probability of

treated units anticipating is bounded by a bounding parameter. Based on our motivating model later, our paper accommodates more than two heterogeneous types with varying discounted parameters. and our identification assumption is also weaker compared to Gong’s because it is implied by his. Moreover, the bounds he provides depend on the signs of treatment effects and anticipation effects, whereas ours do not depend on any unknown parameter of interest. Requiring researchers to know the direction of both effects beforehand is restrictive. For example, in the Medicare Part D case (Alpert, 2016) mentioned before, anticipation effects could be positive if income effects dominate or could be negative if intertemporal substitution effects dominate. Either direction is possible, and our bounds do not require researchers to know the signs of these effects in advance.

Outline The rest of this paper is organized as follows: Section 2 presents our main identification result. Section 3 discusses estimation and inference. Section 4 introduces the bounds which are robust to misspecification. Section 5 extends the analysis to multiple treatment periods setting. Section 6 applies our framework to some empirical settings. Section 7 concludes. Proofs as well as additional results are reported in the Appendix.

2 Analytical Framework and Identification Results

In this section, we introduce the setup and main identification results. We first provide the identified set for the ATT under the canonical DID model with two periods, then we tighten this identified set by including more than one pre-treatment period. A motivating model and sufficient conditions for identification assumptions are provided in subsection 2.2.1.

2.1 Canonical Two Periods Model

We first consider the standard canonical difference-in-differences model

$$Y_t = Y_t(1)D + Y_t(0)(1 - D) \quad \text{for } t = 0, 1 \quad (2.1)$$

where the vector (Y_0, Y_1, D) represents the observed data, while the vector $\{(Y_t(0), Y_t(1))\}_{t=0,1}$ is latent. In this model, $D \in \{0, 1\}$ is the observed treatment that occurs between period 0 and 1. $Y_0, Y_1 \in \mathcal{Y}$ are the observed outcomes in the baseline and follow-up periods, respectively. $Y_t(0)$ and $Y_t(1)$ are potential outcomes that would have been observed in period t had the treatment been externally set to 0 and 1, respectively.

As is often the case, in this paper, we are interested in identifying the average treatment effect on the treated (ATT) which is defined as

$$ATT \equiv \mathbb{E}[Y_1(1) - Y_1(0) \mid D = 1]$$

Assumption 1 (Parallel Trends).

$$SB_0 = SB_1$$

where $SB_t \equiv \mathbb{E}[Y_t(0) \mid D = 1] - \mathbb{E}[Y_t(0) \mid D = 0]$ for $t \in \{0, 1\}$.

We follow Heckman et al. (1998) to interpret the parallel trends assumption in terms of the notion of selection bias. SB is short for selection bias, and the selection bias SB_t is defined as the mean difference of the untreated potential outcome between the treatment and control groups at period t . The parallel trends assumption allows selection on unobservables, but it also restricts this selection since it rules out selection based on time-varying unobservables. Therefore, if only time-invariant unobservables drive selection into treatment, Assumption 1 will hold.

Define the (average) anticipation effect on the treated group as

$$AE \equiv \mathbb{E}[Y_0(1) - Y_0(0) \mid D = 1]$$

and define the OLS estimand and the standard DID estimand as

$$\begin{aligned}\theta_0^{OLS} &= \mathbb{E}[Y_1 \mid D = 1] - \mathbb{E}[Y_1 \mid D = 0] \\ \theta_1^{OLS} &= \mathbb{E}[Y_0 \mid D = 1] - \mathbb{E}[Y_0 \mid D = 0] \\ \theta^{DID} &= \theta_1^{OLS} - \theta_0^{OLS}\end{aligned}$$

Lemma 1. *Under Assumption 1, the standard DID estimand identifies the ATT net of the AE, i.e.,*

$$\theta^{DID} = ATT - AE \tag{2.2}$$

According to Lemma 1, under the parallel trends assumption, the bias of the standard DID estimator is equal to $-AE$, as illustrated in Figure 1. If AE is negative, then θ^{DID} overestimates the ATT . If AE is positive, then θ^{DID} underestimates the ATT . If $AE = ATT \neq 0$, then θ^{DID} is 0 regardless of what the true ATT is, and this “bad case” is shown in Example 1.

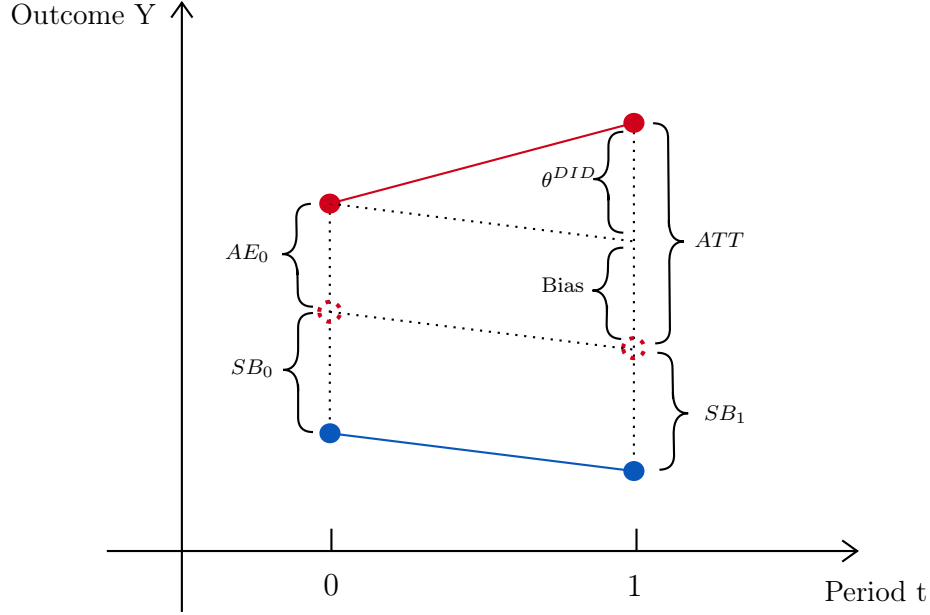


Figure 1: Bias of the standard DID estimand in the presence of anticipation

Notes: Treatment happens between period 0 and period 1. The red solid dot represents the observed average outcome for the treatment group, and the red circle is the potential outcome for the treatment group if treatment never happens. The blue solid dot represents the observed outcome for the control group.

Example 1. For individual i and time t , Y_{it} is the observed outcome, D_i is the treatment indicator, U_{it} is the heterogeneous treatment effect, V_i is the individual-specific unobservable that drives selection into treatment, and ε_{it} is an independent error term. The data generating process (DGP) is as follows:

$$\begin{cases} Y_{it} = U_{it}D_i + V_i + \varepsilon_{it} \\ D_i = \mathbb{1}\{V_i > 0\} \end{cases}$$

where $U_{it} \sim \mathcal{U}(0, 4)$, $V_i \sim \mathcal{N}(0, 1)$, $\varepsilon_{it} \sim \mathcal{N}(0, 2)$ and $\forall t$, $U_{it} \perp\!\!\!\perp V_i$. In this example, the true ATT is equal to 2, and the anticipation effect is also equal to 2, so the standard DID estimand will yield 0 by Lemma 1. Therefore, the standard DID estimand fails to identify the ATT, and the bias arises because the anticipation effect is as large as the ATT.¹

Hence, to identify the ATT, we need to impose some restrictions on AE . Assumption 2 below will connect AE and ATT in some way, which enables us to identify ATT .

¹In this scenario, our DID framework introduced later cannot identify the ATT either. To address this issue, a possible solution is to use the instrumental variable (IV). For example, $Y_{i,t-1}$ can be used as an IV. Note that $Y_{i,t-1}$ is an imperfect IV since the exogeneity assumption is violated, but we can apply the method proposed in Nevo and Rosen (2012) to partially identify ATT.

Assumption 2 (Bounded anticipation effect). *For some known $c \in [0, 1)$,*

$$|AE| \leq c \cdot |ATT|$$

Assumption 2 is a weaker version of the no-anticipation effects, and this kind of assumption is called the bounded-variation assumption in Manski and Pepper (2018). Note that we allow for anticipation, but we still impose some restrictions on it. Specifically, the average anticipation effect is bounded in magnitude by the true ATT of interest. If $c = 0$, then Assumption 2 collapses to the no-anticipation effects assumption that is commonly invoked in the DID literature.

Proposition 1. *Under Assumption 1 and 2, bound on ATT is given by*

$$ATT \in \left[\frac{\theta^{DID}}{1+c}, \frac{\theta^{DID}}{1-c} \right] \cup \left[\frac{\theta^{DID}}{1-c}, \frac{\theta^{DID}}{1+c} \right]. \quad (2.3)$$

Proposition 1 says that ATT will fall within a union bound in the presence of anticipation. This bound depends on a known bounding parameter c , and this known parameter is used to correct the bias in both directions. One could interpret c as a discount factor, which will be mentioned in subsection 2.2.1. The choice of c can play a role in sensitivity analysis, and we can also use empirical literature to calibrate c and assess the sensitivity of the proposed bounds. Furthermore, we will show that when multiple pre-treatment periods are available, data can tell us which value of c is consistent with our model.

2.2 Multiple Pre-Treatment Periods Model

Here, we assume that the researcher has access to baseline information set $\mathcal{T}_0$, which is composed of multiple pre-treatment periods, $\mathcal{T}_0 = \{-t_0, -t_0 + 1, \dots, -1, 0\}$. Define for every $t_0 \in \mathcal{T}_0$,

$$\begin{aligned} \theta_{t_0}^{OLS} &\equiv \mathbb{E}[Y_{t_0} \mid D = 1] - \mathbb{E}[Y_{t_0} \mid D = 0] \\ SB_{t_0} &\equiv \mathbb{E}[Y_{t_0}(0) \mid D = 1] - \mathbb{E}[Y_{t_0}(0) \mid D = 0] \\ AE_{t_0} &\equiv \mathbb{E}[Y_{t_0}(1) - Y_{t_0}(0) \mid D = 1] \end{aligned}$$

Then, we have $\theta_{t_0}^{OLS} = AE_{t_0} + SB_{t_0}$.

Assumption 3 (Strong Parallel Trends).

$$SB_{t_0} = SB_{t'_0} = SB_1 \text{ for all } t_0, t'_0 \in \mathcal{T}_0$$

Assumption 4 (Exponential c-bounded anticipation effects). *For some known $c \in [0, 1)$,*

$$|AE_{t_0}| \leq c^{|t_0|+1} \cdot |ATT| \text{ for all } t_0 \in \mathcal{T}_0$$

Assumption 4 states that individuals possess rational expectations regarding the true ATT, and the anticipation effects on average are discounted from ATT by a parameter c . This assumption can be justified by the observation that at some point (unknown to econometricians) before the treatment, only a small number of people are aware of upcoming intervention or policy, they might respond to that, suggesting that on average the “prior treatment effects” should not be significantly large initially. However, as information spreads exponentially, more and more people are informed, and these effects should grow substantially as well. Therefore, when viewed in reverse, the average anticipation effects are expected to diminish exponentially as one moves further away from the treatment period, and we can also think of them as “discounted treatment effects”. Alternative justification of assumption 4 is discussed in subsection 2.2.1 and a sufficient condition is provided.

Remark 1. Assumption 4 is similar to the exponential discounting model in Malani and Reif (2015) where they assume the anticipation effect is equal to discounted ATT if there is only one post-treatment period, while our assumption only states that the magnitude of AE is bounded by the magnitude of discounted ATT.

2.2.1 A Motivating Model

To motivate Assumption 4 above, let us consider the following forward-looking model (Chow, 1989; Malani and Reif, 2015) in potential outcome framework: for $t \in \mathcal{T}$,

$$Y_t(\mathbf{D}) = Y_t(\mathbf{0}) + \tau_t D_t + \sum_{j=1}^{\infty} \beta^j \mathbb{E}_t(\tau_{t+j} D_{t+j}) \quad (2.4)$$

where $\mathbf{D} \equiv (\dots, D_{t-1}, D_t, D_{t+1}, \dots)$ is the treatment path, $\tau_t \equiv Y_t(\dots, 0, D_t = 1, 0, \dots) - Y_t(\mathbf{0})$ is the contemporaneous treatment effect for each individual in period t , $\beta \in [0, 1)$ is a stochastic discount parameter which is random across individuals.

Assume rational expectation and let expectation be a function of treatment so that

$$\mathbb{E}_t(\tau_{t+j} D_{t+j}) = \tau_{t+j} D_{t+j} + e_{t,t+j} \quad (\text{ or } \tau_{t+j} D_{t+j} = \mathbb{E}_t(\tau_{t+j} D_{t+j}) + e_{t,t+j})$$

where $e_{t,t+j}$ is the forecast error resulting from an agent’s time t forecast of $\tau_{t+j} D_{t+j}$ and $e_{t,t+j}$ is independent with mean 0.

Suppose people can only be treated at $t=1$, it follows that

$$ATT_1 = \mathbb{E}[\tau_1 \mid (\cdots, D_0, D_1, D_2, \cdots) = (\cdots, 0, 1, 0, \cdots)]$$

and for $t_0 \in \mathcal{T}_0 = \{\dots, -1, 0\}$,

$$AE_{t_0} = \mathbb{E}[Y_{t_0}(\dots, D_0 = 0, D_1 = 1, D_2 = 0, \dots) - Y_{t_0}(\mathbf{0}) \mid (\cdots, D_0, D_1, D_2, \cdots) = (\cdots, 0, 1, 0, \cdots)]$$

Proposition 2. *Suppose individuals can only receive treatment in period 1. Under the forward-looking model 2.4, if either of the assumptions below holds, then there exists a constant $c \in [0, 1)$ s.t. $|AE_{t_0}| \leq c^{|t_0|+1} \cdot |ATT_1|$ for all $t_0 \in \mathcal{T}_0$.*

(i) β is constant

(ii) $\beta \perp\!\!\!\perp (Y_1(0, \dots, 0, D_1 = 1, 0, \dots), Y_1(\mathbf{0})) \mid \mathbf{D}$, and β is bounded away from 1.

Remark 2. For two periods $\mathcal{T} = \{0, 1\}$, if β is random but not correlated with potential outcomes $Y_1(0)$ and $Y_1(1)$ for treated people, then $|AE_{t_0}| \leq c \cdot |ATT|$ where c could be the conditional expected value of β for treated people, i.e. $\mathbb{E}(\beta \mid D = 1)$.

Remark 3. The model 2.4 accommodates heterogeneous types of people. Individuals exhibit heterogeneity based on the discount parameter β . An individual i is a non-anticipator if $\beta_i = 0$, and an anticipator if $\beta_i > 0$. A higher value of β indicates a greater degree of forward-looking behavior.

2.2.2 Simple example where Assumption 4 holds

Example 2. Suppose for each individual $i \in \{1, 2, \dots, N\}$ and $t \in \mathcal{T} = \{-T_0, -T_0 + 1, \dots, -2, -1, 0, 1\}$,

$$\begin{cases} Y_{it} = \delta^{|t-1|} U_{it} D_i + V_i + \varepsilon_{it} \\ D_i = \mathbb{1}\{V_i > 0\} \end{cases}$$

where $\delta \in (0, 1)$, $U_{it} \sim \mathcal{U}(0, 4)$, $V_i \sim \mathcal{N}(0, 1)$, $\varepsilon_{it} \sim \mathcal{N}(0, 2)$ and $\forall t, U_{it} \perp\!\!\!\perp V_i$.

In Example 2, all the variables are interpreted in the same manner as in Example 1. The individual-specific variable V_i drives selection into treatment, and U_{it} is the treatment effect which is uniformly distributed and heterogeneous across individuals and time, and the true ATT is equal to 2. However, the magnitude of anticipation effect is governed by a positive constant δ since $AE_t = \delta^{|t-1|} \cdot \mathbb{E}(1 + U_{it})$ and $ATT = \mathbb{E}(1 + U_{i1})$.

Let $\delta = 0.5$, $N = 2000$, $T_0 = 6$. We can easily verify that this DGP satisfies Assumption 3. Furthermore, if we set the bounding parameter $c \geq \delta$, then Assumption 4 will hold. The

average observed outcomes over time for the two groups and the potential outcomes for the treatment group are illustrated in Figure 2, and our model is correctly specified when we set $c \geq 0.5$.

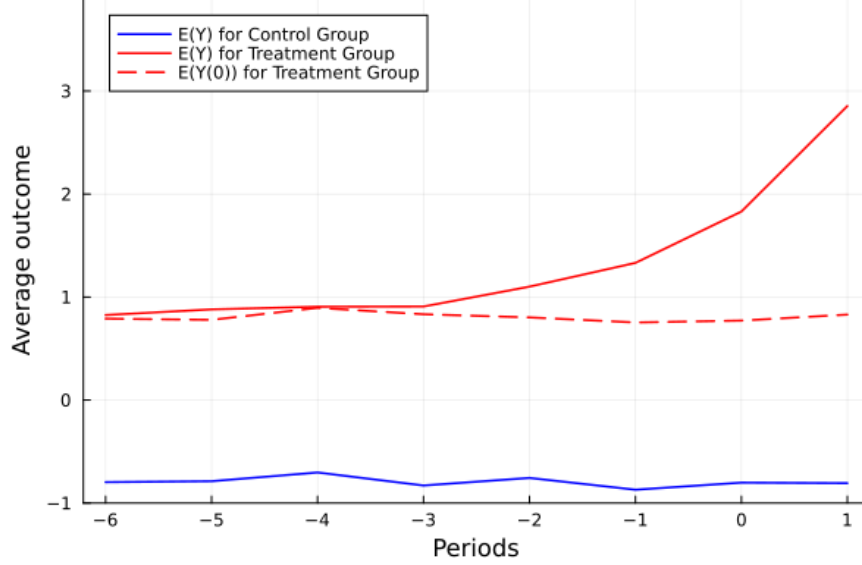


Figure 2: Average outcomes over time for two groups

Notes: Treatment happens between period 0 and period 1. The red solid line represents the observed average outcome for the treatment group, and the red dashed line is the potential outcome for the treatment group if treatment never happens. The blue solid line represents the observed outcome for the control group. In period 1, the difference between the red solid line and the red dashed line is ATT , and the difference between the red dashed line and the blue solid line is SB_1 . In period 0, the difference between the red solid line and the red dashed line is AE_0 , and the difference between the red dashed line and the blue solid line is SB_0 . Similarly, during the pre-treatment periods, the difference between the red solid line and the blue solid line can be decomposed into corresponding AE and SB .

2.2.3 Identification in multiple pre-treatment periods model

By using a large information set with the assumptions above, we can refine the bounds on ATT . Specifically, in this case, we could use information from multiple pre-treatment periods to tighten our bounds.

Proposition 3. *Under Assumptions 3 and 4, bound on ATT is given by*

$$ATT \in \Theta_{ATT} \equiv \left[\max_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\}, \min_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\} \right] \cup \left[\max_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\}, \min_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\} \right] \quad (2.5)$$

where $\theta_{t_0}^{DID} \equiv \theta_1^{OLS} - \theta_{t_0}^{OLS}$.

The bounds in (2.5) take the form of union of two intersection bounds, and they are tighter than the bounds in (2.3) under two periods. By plotting the bounds as a function of c , we can quantify the impact of failures of no-anticipation effects on our knowledge about ATT .

3 Estimation and Inference

In this section, we will show how to estimate and conduct inferences for our bound from Proposition 3. Since our bound takes the form of intersection-union bounds, it can be implemented easily using existing methods.

3.1 Estimation

For estimation, we can simply point estimate Θ_{ATT} by:

$$\hat{\Theta}_{ATT} \equiv \left[\max_{t_0 \in \mathcal{T}_0} \left\{ \frac{\hat{\theta}_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\}, \min_{t_0 \in \mathcal{T}_0} \left\{ \frac{\hat{\theta}_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\} \right] \cup \left[\max_{t_0 \in \mathcal{T}_0} \left\{ \frac{\hat{\theta}_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\}, \min_{t_0 \in \mathcal{T}_0} \left\{ \frac{\hat{\theta}_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\} \right]$$

where $\hat{\theta}_{t_0}^{DID}$ is the standard DID estimator using period 0 and period t_0 .

Since $\max_{t_0 \in \mathcal{T}_0} \left\{ \frac{\hat{\theta}_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\}$, $\min_{t_0 \in \mathcal{T}_0} \left\{ \frac{\hat{\theta}_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\}$ and the other two estimators are consistent with the upper and lower boundary of each interval, it is easy to show $\hat{\Theta}_{ATT}$ is consistent to Θ_{ATT} in Hausdorff distance.

If we merely point estimate the bound without applying any inferential procedure, estimated bounds will lead to an issue of undercoverage. Therefore, we must resort to employing some inferential procedures.

3.2 Inferential Procedure

For inference, our identified set in Proposition 3 takes the form of a union of two intersection bounds, so we employ a variant of the inferential procedure proposed by [Nevo and Rosen \(2012\)](#) to construct valid confidence bounds. The procedure is shown as follows:

1. Fix c , obtain $\hat{\Theta}_{ATT} = [\max\{L_{n,t_0}\}, \min\{U_{n,t_0}\}] \cup [\max\{U_{n,t_0}\}, \min\{L_{n,t_0}\}]$

where L_{n,t_0} is consistent estimator of $\frac{\theta_{t_0}^{DID}}{1+c^{|t_0|+1}}$ and U_{n,t_0} is consistent estimator of $\frac{\theta_{t_0}^{DID}}{1-c^{|t_0|+1}}$, $\forall t_0 \in \mathcal{T}_0$

2. For the first interval of Θ_{ATT} : $[\max\{L_{n,t_0}\}, \min\{U_{n,t_0}\}]$
 Let $\bar{\sigma}_l \equiv \max_{t_0 \in \mathcal{T}_0} \hat{s}_{l,t_0}$, $\bar{\sigma}_u \equiv \max_{t_0 \in \mathcal{T}_0} \hat{s}_{u,t_0}$, where $\hat{s}_{l,t_0}$, $\hat{s}_{u,t_0}$ are the standard errors of the estimates L_{n,t_0} , U_{n,t_0} , respectively.
 Let $L_n^* \equiv \max_{t_0 \in \mathcal{T}_0} \{L_{n,t_0}\}$, $U_n^* \equiv \max_{t_0 \in \mathcal{T}_0} \{U_{n,t_0}\}$.
 Compute index set $\mathcal{J}_l^* = \{t_0 \in \mathcal{T}_0 : L_{n,t_0} \geq L_n^* - 2\bar{\sigma}_l \sqrt{\log n}\}$ and $\mathcal{J}_u^* = \{t_0 \in \mathcal{T}_0 : U_{n,t_0} \leq U_n^* + 2\bar{\sigma}_u \sqrt{\log n}\}$
3. Define $\Delta_n \equiv |U_n^* - L_n^*|_+$ and $p_n \equiv 1 - \Phi(\log n \times \Delta_n)\alpha$
 Let $\hat{\Omega}_l$ and $\hat{\Omega}_u$ be consistent estimators for the covariance of $\{L_{n,t_0} : j \in \mathcal{J}_l^*\}$ and $\{U_{n,t_0} : j \in \mathcal{J}_u^*\}$, respectively. (Such estimates may be obtained either by analytical calculation of asymptotic variance formulas or by bootstrapping)
 Let Z^l and Z^u denote $|\mathcal{J}_l^*|$ and $|\mathcal{J}_u^*|$ mean-zero multivariate normal random variables with variances $\hat{\Omega}_l$ and $\hat{\Omega}_u$.
 Compute the p_n quantiles of $\max\{Z^l\}$ and $\max\{Z^u\}$, denoted $q^l(p_n)$ and $q^u(p_n)$.
4. A $1 - \alpha$ confidence interval for the first interval of Θ_{ATT} is the given by $CI_{1-\alpha}^1 = [L_{1-\alpha}, U_{1-\alpha}]$ where
 $L_{1-\alpha} \equiv \max_{t_0 \in \mathcal{J}_l^*} \{L_{n,t_0} - q^l(p_n)\}$
 $U_{1-\alpha} \equiv \min_{t_0 \in \mathcal{J}_u^*} \{U_{n,t_0} + q^u(p_n)\}$
5. For the second interval of Θ_{ATT} , exchange L_{n,t_0} and U_{n,t_0} , then repeat the step 2 - 4 above and obtain $CI_{1-\alpha}^2$.
6. Take the union of $CI_{1-\alpha}^1$ and $CI_{1-\alpha}^2$, and get the $1 - \alpha$ confidence interval for the parameter of interest, ATT , for given c .

This confidence set $CI_{1-\alpha}^1$ provides an asymptotic confidence interval for the bounded parameter of interest, which is uniformly valid over points in the identified set when true ATT is positive. It is the same for $CI_{1-\alpha}^2$ when true ATT is negative. Since we take the union of $CI_{1-\alpha}^1$ and $CI_{1-\alpha}^2$ in the end, our confidence set should be a little bit conservative, but it is still valid. Later, we probably need to refine our confidence set by using some inferential procedure for union bounds.

Next, we will revisit the DGP in Example 2 to illustrate what our bounds look like.

Example 3. (Cont.) Suppose for each individual $i \in \{1, 2, \dots, N\}$ and $t \in \mathcal{T} = \{-T_0, -T_0 + 1, \dots, -2, -1, 0, 1\}$

$$\begin{cases} Y_{it} = \delta^{|t-1|} U_{it} D_i + V_i + \varepsilon_{it} \\ D_i = \mathbb{1}\{V_i > 0\} \end{cases}$$

where $\delta \in (0, 1)$, $U_{it} \sim \mathcal{U}(0, 4)$, $V_i \sim \mathcal{N}(0, 1)$, $\varepsilon_{it} \sim \mathcal{N}(0, 2)$ and $\forall t, U_{it} \perp V_i$.

Let $\delta = 0.5, N = 2000, T_0 = 6$. In this DGP, the true ATT is equal to 2. Figure 3 compares the standard DID estimate $\hat{\theta}^{DID}$ and the bounds on ATT with considering the anticipation effect under different values of c . If we merely point estimate bounds, as shown in plot (a) of Figure 3, our bounds can cover the true ATT when c is greater than around 0.6. However, by simulation, this naive bound suffers the issue of undercoverage around $c = \delta = 0.5$.² If we construct the confidence band by using Nevo and Rosen's (2012) inferential procedure, we can obtain a 95% confidence band, as shown in plot (b) of Figure 3. The resulting confidence band for bounds on ATT includes the true value if the model is correctly specified ($c > 0.5$). In contrast, the standard 95% confidence interval for θ^{DID} does not contain the true value. By the Monte Carlo simulation, the average of $\hat{\theta}^{DID}$ is around 1, which means $\hat{\theta}^{DID}$ is biased in the presence of the anticipation effects.

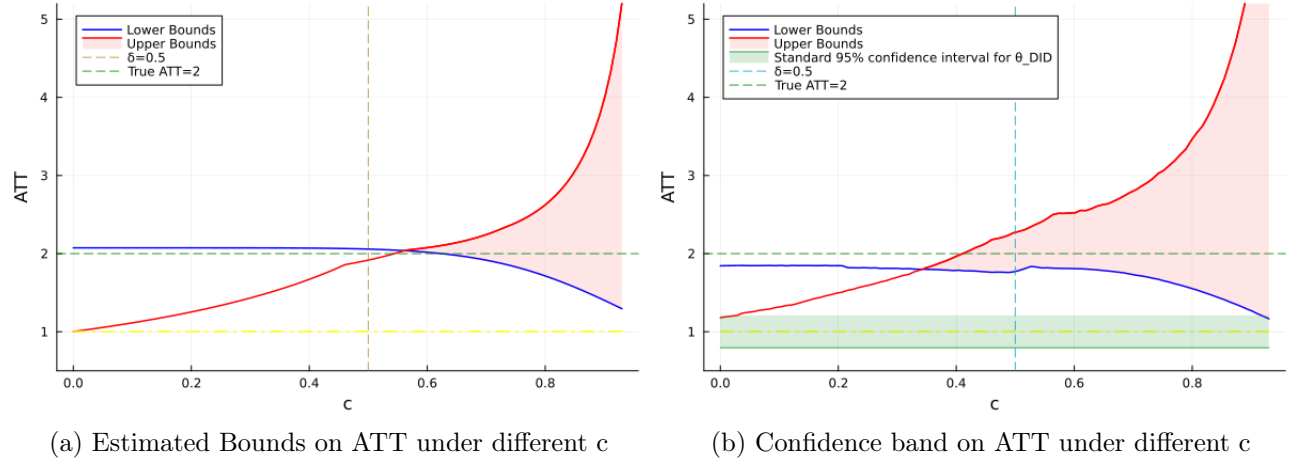


Figure 3: Bounds with and without inferential procedure

Notes: The red line is the upper bound of ATT, and the blue line is the lower bound of ATT. The yellow dashed line is the standard DID estimate, and the green band is the 95% confidence interval for θ^{DID} . The red area indicates that the model is not falsified by the data under corresponding values of c . The vertical line indicates $\delta = 0.5$

²Verification of coverage of bound for different c is shown in the Appendix.

3.3 Breakdown Frontiers

We can find the range for values of the parameter c for which a given conclusion can or cannot be rejected by the model. The literature refers to this set of values as the breakdown frontier (See [Manski and Pepper, 2018](#); [Masten and Poirier, 2021](#)). For example, what values of c are not consistent with the hypothesis $ATT \geq \underline{\theta}$? What values of c are not consistent with the hypothesis $ATT \leq \bar{\theta}$? To answer the first question, we need to find c such that $ATT_{UB} < \underline{\theta}$, and for the second, we need to search for values of c such that $ATT_{LB} > \bar{\theta}$, where ATT_{LB} and ATT_{UB} denote respectively the lower and upper bounds of Θ_{ATT} .

4 Robust DID Bounds

In this section, we derive a misspecification robust bound that is robust to either one of two assumptions being violated. Subsequently, we relax both identifying assumptions together and derive a bound that is robust to both of them being violated.

4.1 Misspecification Robust Bounds

Define the set of assumptions

$$A_1 \equiv \{\text{Strong parallel trends, Exponential c-bounded anticipation}\}$$

$$A_2 \equiv \{\text{No anticipation, Bias set stability}\}$$

$$A \equiv \{A_1, A_2\}$$

where bias set stability assumption is introduced in [Ban and Kedagni \(2022\)](#), which states the following:

$$SB_1 \in [\inf_{t_0 \in \mathcal{T}_0} SB_{t_0}, \sup_{t_0 \in \mathcal{T}_0} SB_{t_0}]$$

Let $\Theta(A_j)$ denote the identified set of the given assumption A_j . Using the concept of misspecification robust bound introduced in [Kédagni, Li, and Mourifié \(2020\)](#), we can show that the misspecification has a robust bound for the set of assumptions A is

$$\Theta^*(A) \equiv \begin{cases} \Theta(A) = \Theta(A_1 \cup A_2) = \{\theta^{DID}\} & \text{if } \theta_0^{OLS} = \theta_{t_0}^{OLS} \text{ for all } t_0 \in \mathcal{T}_0 \\ \Theta(A_1) \cup \Theta(A_2) & \text{otherwise} \end{cases} \quad (4.1)$$

where

$$\Theta(A_1) = \left[\max_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\}, \min_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\} \right] \cup \left[\max_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\}, \min_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\} \right] \quad (4.2)$$

and

$$\Theta(A_2) = \left[\theta_1^{OLS} - \sup_{t_0 \in \mathcal{T}_0} SB_{t_0}, \theta_1^{OLS} - \inf_{t_0 \in \mathcal{T}_0} SB_{t_0} \right] \quad (4.3)$$

Equation (4.3) is from Proposition 1 in [Ban and Kedagni \(2022\)](#). Note that $A_1 \cup A_2 = \{\text{Strong parallel trends, No anticipation}\}$, because no anticipation implies exponential c-bounded anticipation and strong parallel trends imply bias set stability.

4.2 Relaxing Both Parallel Trends and No-anticipation Effects

Assumption 5 (Bias set stability). *Let $\mathcal{T}_0$ be the index set of pre-treatment periods.*

$$SB_1 \in \left[\min_{t_0 \in \mathcal{T}_0} SB_{t_0}, \max_{t_0 \in \mathcal{T}_0} SB_{t_0} \right]$$

Proposition 4. *Under Assumption 4 and 5, bound on the ATT is given by*

$$ATT \in \Theta_{ATT} \equiv \left[\min_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\}, \max_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\} \right] \cup \left[\min_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\}, \max_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\} \right] \quad (4.4)$$

where $\theta_{t_0}^{DID} \equiv \theta_1^{OLS} - \theta_{t_0}^{OLS}$.

Now, the bounds in (4.4) take the form of a union bound. It is wider than the bounds before, but it is robust to the violation of two identifying assumptions at the same time. For inference, we have to use the inferential procedure for the union bounds.

5 Extension to Multiple Treatment Periods

In this section, we extend our assumptions and framework to the multiple treatment periods setting, characterized by a single pre-treatment period followed by multiple post-treatment periods. Within this context, there exist many groups of people, and each group can be

treated at different times. We develop a testable implication of two extended assumptions and provide an identification result that allows for anticipation in the staggered adoption design.

The model is as follows:

$$Y_t = \sum_{(d_1, \dots, d_T) \in \{0,1\}^T} Y_t(0, d_1, \dots, d_T) \mathbb{1}\{D_0 = 0, D_1 = d_1, \dots, D_T = d_T\}, \quad t = 0, \dots, T \quad (5.1)$$

where Y_t denotes the observed outcome at period t , and D_t is the observed treatment status at period t with $D_0 = 0$ by definition, while $Y_t(0, d_1, \dots, d_T)$ is the potential outcome when the treatment path $(D_0, D_1, \dots, D_T)$ is externally set to $(0, d_1, \dots, d_T)$.

5.1 Testable Implication for Extended PT and No-AE

Assumption 6 (Extended Parallel Trends).

$$\mathbb{E}[Y_t(0, \dots, 0) - Y_0(0, \dots, 0) \mid (D_0, D_1, \dots, D_T) = (0, d_1, \dots, d_T)] = \mathbb{E}[Y_t(0, \dots, 0) - Y_0(0, \dots, 0)]$$

for all t and all treatment path $(0, d_1, \dots, d_T)$.

Assumption 7 (Extended no-anticipation effects).

$$\mathbb{E}[Y_t(0, \dots, 0, d'_{t+1}, \dots, d'_T) - Y_t(0, \dots, 0) \mid (D_0, D_1, \dots, D_T) = (0, d_1, \dots, d_T)] = 0$$

for all t , $(0, d_1, \dots, d_T)$ and $(0, \dots, 0, d'_{t+1}, \dots, d'_T)$.

Assumption 6 states that the average outcomes of all groups at any time would have followed the same trend in the absence of treatment. Assumption 7 states that for any group of people, if they have not been treated by period t , no matter what happens later, the average potential outcome at t is the same as the average potential outcome of never being treated. In other words, if individuals have not received treatment before period t , they will not expect any treatment in the future to have an impact on the current situation. In this case, the future treatment does not affect the current outcome. However, if individuals have been treated before period t , they might have anticipation effects if this happens again in the future, then the future treatment will potentially affect the current outcome.

Proposition 5. Under Assumption 6 and 7, the following must hold:

$$\mathbb{E}[Y_t - Y_0 \mid (D_0, D_1, \dots, D_T) = (0, \dots, 0, d_{t+1}, \dots, d_T)] = \mathbb{E}[Y_t - Y_0 \mid (D_0, D_1, \dots, D_T) = (0, \dots, 0)] \quad (5.2)$$

for all t and $(0, \dots, 0, d_{t+1}, \dots, d_T)$.

Proposition 5 implies that all “not-yet-treated” groups at period t , including the never-treated, have equal (linear) trends between periods 0 and t . The implication (5.2) is testable since all variables and groups involved are observed.

Remark 4. In the staggered adoption design, we propose the following test procedure. Suppose we observe T treatment periods. Define

$$D^g \equiv \mathbb{1}\{(D_0, D_1, \dots, D_g, \dots, D_T) = (0, 0, \dots, d_g = 1, \dots, 1)\}$$

$$D^0 \equiv \mathbb{1}\{(D_0, D_1, \dots, D_g, \dots, D_T) = (0, \dots, 0)\}$$

$$\theta_s^g \equiv (\mathbb{E}[Y_{it} \mid D_i^g = 1, t = s] - \mathbb{E}[Y_{it} \mid D_i^0 = 1, t = s]) - (\mathbb{E}[Y_{it} \mid D_i^g = 1, t = 0] - \mathbb{E}[Y_{it} \mid D_i^0 = 1, t = 0])$$

where g refers to the first time being treated, i refers to individual i . $D_i^g = 1$ if individual i is treated at period g for the first time, and we say individual i is in group g . Then, we can run the following regression (TWFE approach) to obtain the parameter θ_s^g (See Wooldridge, 2021; Ban and Kedagni, 2022): For $t \in \{0, 1, \dots, T\}$,

$$Y_{it} = \beta + \sum_{g=1}^T \gamma^g D_i^g + \sum_{s=1}^T \delta_s \mathbb{1}\{t = s\} + \sum_{s=1}^T \sum_{g=1}^T \theta_s^g D_i^g \mathbb{1}\{t = s\} + \varepsilon_{it} \quad (5.3)$$

By Proposition 5, if Assumptions 6 and 7 hold, $\theta_s^g = 0$ when $g > s$. Therefore, we can test whether $\theta_s^g = 0$ under the null hypothesis that Assumption 6 and Assumption 7 are satisfied, if there exists a g and s such that $g > s$ and $\theta_s^g \neq 0$, then it implies at least one assumption is violated. Naturally, we are able to implement the Wald-test or F-test to assess the validity of these assumptions. One simulation example and the verification of hypothesis testing are provided in Appendix B.2.

5.2 Identification under a Relaxed Assumption in Staggered Adoption Design

In the setting of the staggered adoption design, if individuals are allowed to anticipate after the first time the initial treated group receives treatment, Ban and Kedagni (2022) shows that the estimation equation (5.3) remains robust to the presence of anticipation. But we can relax further, allowing people to anticipate even earlier.³ In this case, the TWFE estimate would be biased.

³It is plausible that people begin to anticipate after observing other groups being treated. However, it is also possible that some people change their behavior even before the first treatment of other groups. This could be because of having inside information, being influenced by the congressional negotiation process, or exposed to some rumors.

Let $\mathcal{T} = \{0, 1, 2, \dots, T_1\}$. For each group $s \in \mathcal{T} \setminus \{0\}$ (the group of people whose first treatment is in period s), we define

$$\begin{aligned} ATT_t^s &\equiv \mathbb{E}[Y_t(0, \dots, 0, d_s = 1, \dots, 1) - Y_t(0, \dots, 0) \mid (D_0, D_1, \dots, D_T) = (0, \dots, d_s = 1, \dots, 1)] \\ AE_t^s &\equiv \mathbb{E}[Y_{t_0}(0, \dots, 0, d_s = 1, \dots, 1) - Y_{t_0}(0, \dots, 0) \mid (D_0, D_1, \dots, D_T) = (0, \dots, d_s = 1, \dots, 1)] \end{aligned}$$

Assumption 8 (Extended bounded anticipation effects). *For some known $c \in [0, 1)$, given any group $s \in \mathcal{T} \setminus \{0\}$, for all $t_0 < s$,*

$$|AE_{t_0}^s| \leq c^{s-t_0} |ATT_s^s|$$

Assumption 8 is also motivated by the model discussed in subsection 2.2.1. We assume that for group s , people will respond solely to the ATT at period s in advance. This is because, by definition, ATT_s^s already incorporates information including both the contemporaneous effect and the expected future treatment effects under the forward-looking model. Additionally, we restrict the magnitude of anticipation effects is bounded by discounted ATT_s^s .

Lemma 2. *In the staggered adoption design, Assumption 6 is equivalent to the following:*

$$SB_t^s = SB_0^s \text{ for all } t, s$$

where

$$\begin{aligned} SB_t^s &\equiv \mathbb{E}[Y_t(0, \dots, 0) \mid (D_0, D_1, \dots, D_T) = (0, \dots, d_s = 1, \dots, 1)] \\ &\quad - \mathbb{E}[Y_t(0, \dots, 0) \mid (D_0, D_1, \dots, D_T) = (0, \dots, 0)] \end{aligned}$$

Lemma 2 above shows that the extended parallel trends assumption is equivalent to the stability of selection bias, a perspective consistently used throughout our paper.

Given group $s \in \mathcal{T} \setminus \{0\}$, for any $t, t' \in \mathcal{T}$, define OLS and DID estimands as follows:

$$\begin{aligned} \theta_t^{OLS,s} &\equiv \mathbb{E}[Y_t \mid (D_0, D_1, \dots, D_T) = (0, \dots, d_s = 1, \dots, 1)] - \mathbb{E}[Y_t \mid (D_0, D_1, \dots, D_T) = (0, \dots, 0)] \\ \theta_{t,t'}^{DID,s} &\equiv \theta_t^{OLS,s} - \theta_{t'}^{OLS,s} \end{aligned}$$

Since we are interested in the treatment effects after the intervention is adopted, we will use the assumptions above to identify the ATT_t^s for $t \geq s$. The bounds on ATT_t^s are given by the proposition 6.

Proposition 6. *Under Assumption 6 and 8, for $s \geq t$, bounds on ATT_t^s are given in*

$$ATT_t^s \in \left[\max_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \max_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \right\} \right\}, \min_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \min_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}} \right\} \right\} \right] \cup \left[\max_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \max_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}} \right\} \right\}, \min_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \min_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \right\} \right\} \right] \quad (5.4)$$

In particular,

$$ATT_s^s \in \left[\max_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \right\}, \min_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}} \right\} \right] \cup \left[\max_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}} \right\}, \min_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \right\} \right] \quad (5.5)$$

6 Empirical Illustration

In this section, we apply our framework (the multiple pre-treatment periods setting) to the context examined by [Kresch \(2020\)](#) and [Ladino, Saavedra, and Wiesner \(2021\)](#) (hereafter referred to as “LSW”). Both studies involve situations where future treatment may potentially affect current outcomes. Our framework enables us to account for anticipation effects, enhancing our ability to provide robust inferences for these papers.

6.1 Kresch (2020)

[Kresch \(2020\)](#) studies the effect of policy reform on the investment behavior of municipal companies in the Brazilian water and sanitation (WS) sector. The treatment group is self-run municipalities that provide service through a municipal self-run company, while the control group is state-run municipalities that contract these services out to a state-run company. Self-run companies are exposed to the takeover threat by state-run companies. A reform that clarified the relationship between municipal and state providers in the WS sector was proposed in 2005 and approved by Congress in 2007. [Kresch \(2020\)](#) utilizes this reform as a natural experiment and uses DID to study the effect of the reform on investment decisions. [Kresch \(2020\)](#) uses the year of proposal of the bill, rather than the year of approval, as the treatment period. The original estimation equation in [Kresch \(2020\)](#) is the following

standard TWFE specification with covariates X entering the model linearly:

$$Y_{mt} = \alpha + \gamma_m + \lambda_t + \delta \cdot Reform_{mt} + \beta X_{mt} + \theta \cdot InitialInvest \times timetrend + \varepsilon_{mt} \quad (6.1)$$

where Y_{mt} is the investment level of municipality m in year t , and the data includes 12 years from 2001 to 2012. The reform $Reform_{mt}$ is equal to 1 for self-run municipalities after the legislation was proposed ($t > 2005$). The outcome of interest is the yearly investment at the municipal level. The dataset contains 7 categories of outcomes based on the nature of the investment, as described in Table 1. The covariates X_{mt} include municipality m 's population, gross domestic product (GDP) and taxes, water-intensive industry variables (e.g., agriculture, livestock production), etc.⁴

Table 1: List of Outcome Variables

No.	Name	Classified as..
1	Total Investment	-
2	Investment from Self-financing	Source
3	Investment from Loans and Debt	Source
4	Investment from Government Grants	Source
5	Investment in Water Network	Destination
6	Investment in Sewer Network	Destination
7	Other Network Investments	Destination

Notes: Category (2) (3) (4) are three origins or sources of investment, and (5) (6) (7) are three destinations of investment, so total investment can be calculated as either the sum of “origin” investments or the sum of the “destination” investments.

Anticipation of such reform before treatment is plausible due to the administration’s focus on WS sector improvement following the 2002 national election, as noted by [Heller \(2007\)](#), and we assume the timing of the anticipation is unknown for researchers since it may vary across different units. Some municipalities may be more informed than others and may know about the proposal way before the official proposal date. Additionally, the direction of the anticipation effects is ambiguous. On the one hand, self-run municipalities might anticipate that the risk of the takeover will be eliminated, incentivizing them to invest more.

⁴We also include these covariates in our framework. We first linearly partial out covariates from the outcome variable, then use the residualized outcome to get each estimate nonparametrically. This accounts for the possibility that the change in investment is driven by some confounding factors rather than the reform itself.

On the other hand, state-run companies may expedite taking over the self-run companies before potential future reform, causing self-run companies to be hesitant about investment at the current stage. Given these considerations, our framework allows us to use the year of formal approval (2007) as the treatment period, accounting for anticipation effects before the policy intervention. To fit in our framework, we use 2001-2007 as the pre-treatment periods and 2008 as the post-treatment period.

Figure 4 compares the standard DID estimates⁵ with our bounds on ATT for total investment and other three subcategories (2) (3) (4) under different values of c .⁶ Bounds for other outcomes (5) (6) (7) are shown in Figure B.2 in the Appendix. All of these bounds suggest that the no-anticipation effect is not falsified, which means according to the data, it is possible that there is no anticipation. For total investment, bounds show that ATT is significantly positive if $c < 0.5$, and the significance is lost if $c > 0.5$, while the standard DID estimate is not significantly different from 0. Similarly, the bounds on subcategory (2) (3) are above 0 for small values of c , meaning that there are significant post-reform increases for self-run municipalities in the aspect of “self-financing” as well as “loan and debt” when there is mild anticipation. On the contrary, there is no statistically significant effect on government grants, as the amount of yearly federal grants appears to be distributed equally to both types of companies.

Table 2 compares our bounds with estimates from Kresch (2020) and generalized DID (GDID) bounds from Ban and Kedagni (2022)⁷, although they consider different specifications and use different estimation approaches. The first two columns of table 2 include estimated coefficients and 95% confidence interval for each type of investment in Kresch (2020). These TWFE estimates in Kresch (2020) show that the effects on all categories except government grants are significantly positive. Note that under relatively mild anticipation effects, our bounds are tighter than the confidence intervals of the TWFE estimates, but the sign of the treatment effect of the reform is consistent with the estimated effects given by the TWFE. Therefore, the results in Kresch’s paper appear somehow robust to mild anticipation. The last column is the 95% confidence interval of bounds robust to the violation of parallel trends, and it shows that the confidence intervals for all outcomes contain 0, suggesting that there

⁵The standard DID estimate is obtained by using only two periods. Specifically, we use year 2007 as the pre-treatment period and 2008 as the post-treatment period.

⁶When implementing the inferential procedure shown in Section 3, we estimate standard DID estimands for each period in one single equation, where we include all the time dummies for pre-treatment periods and all their interaction terms with the indicator of treatment group. This enables us to preserve dependence between DID estimators for each period and obtain the variance-covariance matrix of them, which will be used in the inferential procedure

⁷This paper derives the bound which is robust to violation of the parallel trends assumption, and it also includes Kresch (2020) as an example in the application.

is no significant change in investment after the reform. This result is because the violation of the pre-trend is relatively large, which results in a larger bound for ATT.

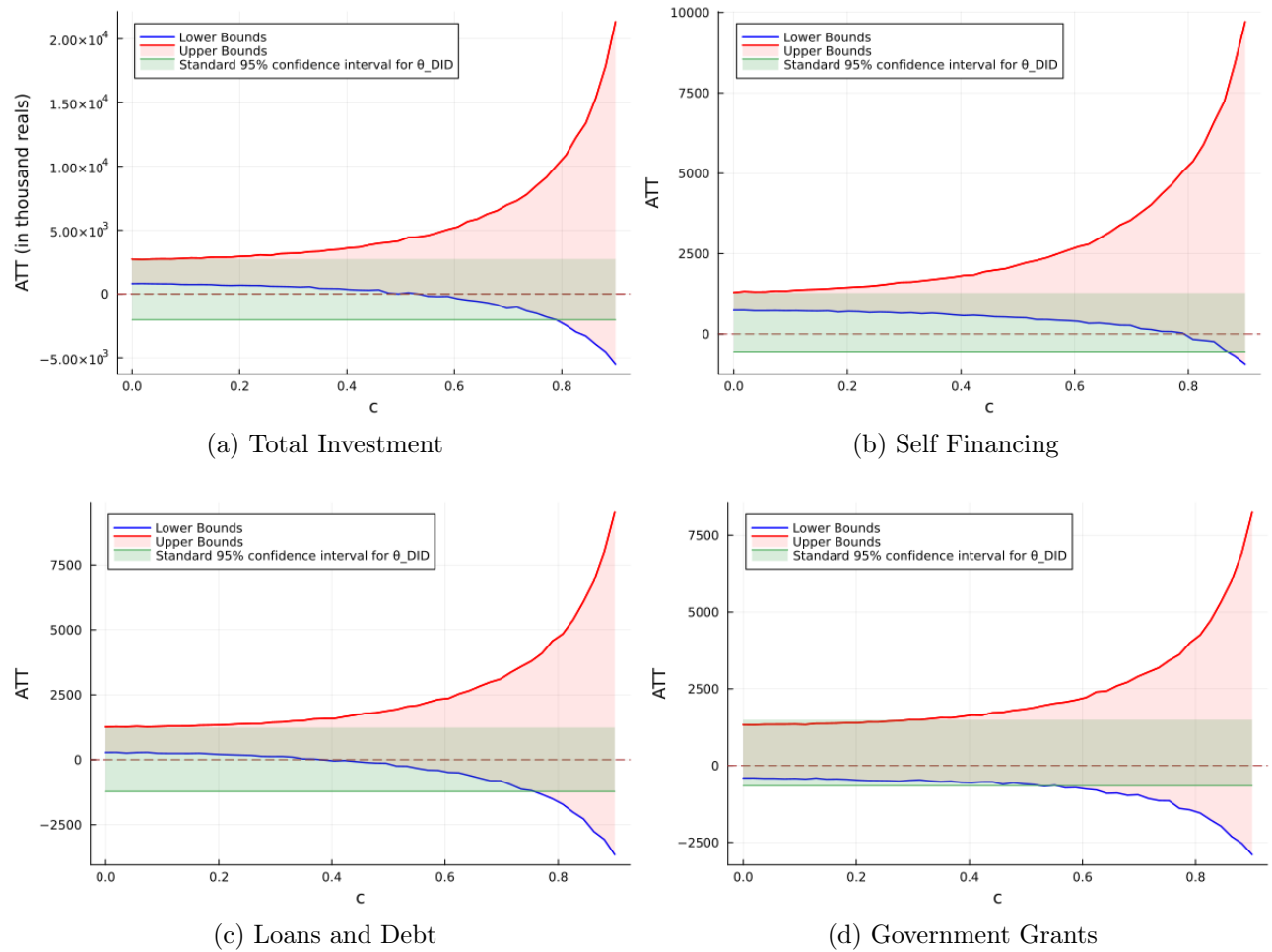


Figure 4: Bounds on ATT for Investments

Table 2: Estimated effects from different specifications

	Kresch		95% CI for bounds robust to anticipation				GDID
	Coef.	95% CI	$c = 0$	$c = 0.3$	$c = 0.6$	$c = 0.9$	
Total Investment	2868	(282,5453)	(784,2690)	(585,3162)	(-398,5244)	(-5804,20673)	(-3979,3452)
Self-Financing	1798	(837,2759)	(730,1302)	(660,1588)	(371,2663)	(-784,9692)	(-620,1951)
Loans and Debt	2124	(319,3929)	(276,1258)	(126,1433)	(-454,2352)	(-3802,9249)	(-626,3013)
Government Grants	-93	(-677,491)	(-414,1326)	(-493,1472)	(-772,2224)	(-2991,7918)	(-1055,523)

6.2 Ladino, Saavedra and Wiesner (2021)

Ladino, Saavedra, and Wiesner (2021) studies the implementation effects of the PNIS program in Colombia, a program of payments for illegal crop substitution that aims to provide incentives for farmers to transition from coca cultivation to legal crops. They compare areas that receive/do not receive payments before/after the program's payments were effectively delivered. The program was announced in 2014 and was implemented in 2017. The equation estimated in their paper is

$$y_{it} = \beta(PNIS_i \times PostPNIS_{it}) + \alpha_i + \lambda_t + \varepsilon_{it}$$

where y_{it} is the hectares planted with coca in grid i during year t . $PostPNIS_{it}$ is a dummy variable that equals 1 after grid i receives eradication payments. α_i, λ_t are grid and time fixed effects.

First, we use the period 2010–2016 as the pre-treatment periods and 2017 as the post-treatment period, using the DID method to assess the policy effectiveness. However, the pre-treatment periods 2010–2016 could be contaminated by anticipation effects, which could bias the standard DID estimate. This is because payments to farmers are conditional on them having a large area of illegal crops. To be eligible for the program, farmers may have an incentive to increase their coca crops before its implementation. Therefore, there might exist anticipation effects, and the direction of these anticipation effects is likely positive, which would overestimate the magnitude of true implementation effect. As shown in Figure 5 (a), the standard two-period DID estimate is significantly negative. Our robust bounds also indicate significantly negative effects, but they correct the downward bias in the presence of anticipation. Moreover, since we use more information (more than 1 pre-treatment period) to construct bounds, the bounds are tighter than the confidence interval of the standard DID when the bounding parameter c is not very large.

Secondly, since the PNIS program rollout in 2017 did not cover all the areas in the treatment group and full coverage was achieved in 2018, we also use the period 2010–2017 as the pre-treatment periods and 2018 as the post-treatment period. Again, the issue of anticipation effects still exists. Units already treated in 2017 may have reduced coca area in 2017, which would be considered anticipation effects due to the choice of treatment year, making the direction of anticipation effects ambiguous. For our analysis, we just assume they are bounded by the true ATT in magnitude. Figure 5 (b) shows that our model is rejected by the data when $c < 0.4$, indicating the presence of anticipation effects that follow an exponential decaying pattern. If $c > 0.4$, the bound shows that the ATT is significantly

negative, consistently less than -0.5 for most values of c . This implies that after the program’s full implementation in 2018, the treated areas reduced coca cultivation by more than 0.5 hectares per 1-kilometer grid on average. In contrast, the 95% confidence interval of the standard DID estimate falls outside our bound for most values of c , and it underestimates the implementation effects in magnitude.

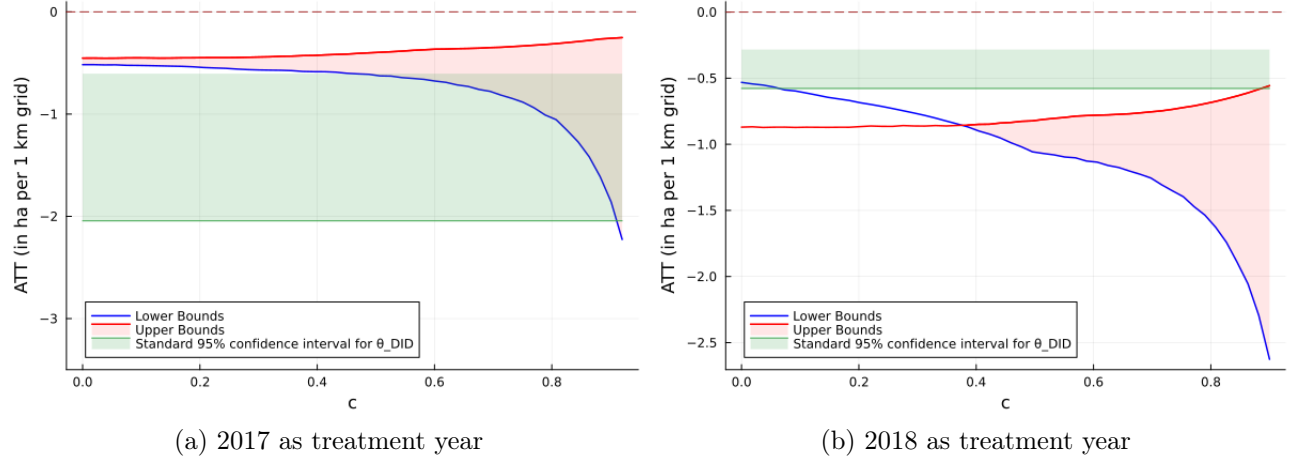


Figure 5: Bounds on ATT for Coca Acreage

Notes: The red line is the upper bound of ATT, and the blue line is the lower bound of ATT. The green band is the 95% confidence interval for θ^{DID} . The standard DID estimate in (a) is obtained by using the data from year 2016 and 2017, while the estimate in (b) is based on year 2017 and 2018.

Discussion: [Ladino, Saavedra, and Wiesner \(2021\)](#) also investigates the announcement effects (unintended consequences) of PNIS. For the announcement effects, they compare areas with/without FARC presence before/after the announcement of the PNIS program.⁸ While for the implementation effects, they compare areas that receive/do not receive payments before/after the program’s payments were effectively delivered. By using the TWFE model with corresponding specification, they find that areas with FARC presence increased coca cultivation on average by 0.54 ha on each 1 km grid square before the program (“anticipation effects”), and areas that receive PNIS payments⁹ reduced coca cultivation on average by 1.04 ha per 1 km grid square (“treatment effects”). This provides empirical evidence that anticipation effects are smaller in magnitude than treatment effects, although the magnitude

⁸The PNIS program was agreed upon by FARC (an armed force in Colombia) and the Colombian government. The FARC has better information regarding the programs, and they have an incentive to inform the farmers about the benefits (such as high profit, and eligibility for the program) of having coca crops since coca crops are the primary funding source for these insurgent groups. Therefore, it is likely for farmers to anticipate when there is the presence of FARC.

⁹Areas that receive PNIS payments have substantially less total area than the areas with FARC presence.

of the treatment effect might be overestimated due to the neglect of anticipation effects in its TWFE specification.

In Figure 6, panel (a) presents the raw data of the average coca area over time, categorized into grids with and without FARC presence. Before the PNIS program, both groups experienced a rise in average coca area. Panel (b) presents the dynamic effects of the announcement relative to the base year 2013. We can observe that after the announcement year (2014) and before the implementation year (2017), the coca area in FARC grids is differentially higher year to year compared to no FARC grids. The pattern of the dynamic of the announcement effects (i.e., anticipation effects before implementation) appears to grow exponentially, which provides evidence for our assumption 4.

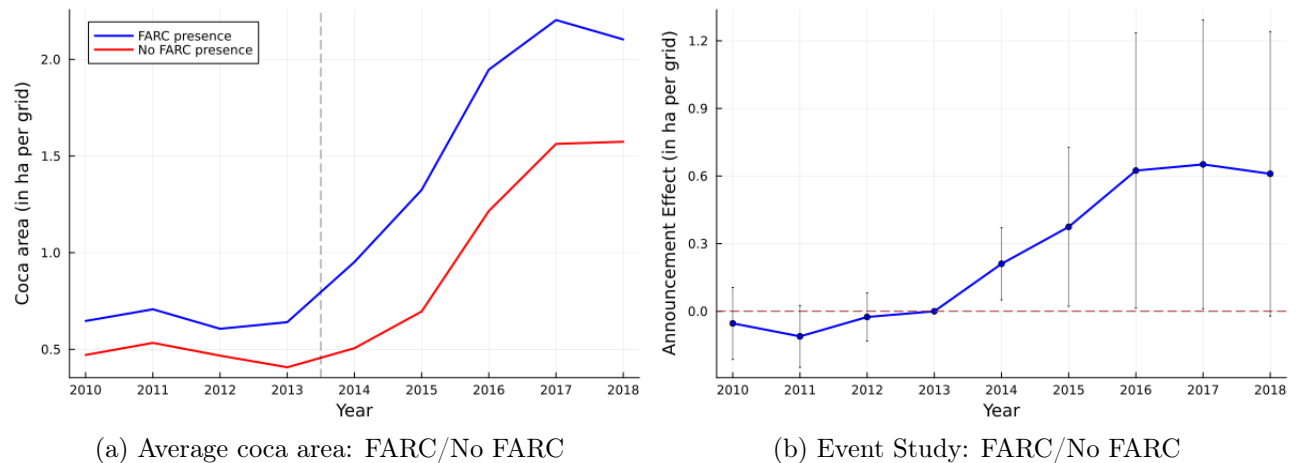


Figure 6: Evolution of coca cultivation with PNIS announcement

Notes: Panel (a) presents the average coca area over time for the grids with and without FARC presence. The blue line represents areas with FARC presence, while the red line represents areas without FARC presence. The vertical line represents the PNIS announcement. Panel (b) illustrates the dynamic effects of the announcement on coca cultivation from the event study analysis. The segment containing each point is the 95% confidence interval for the estimated effect.

7 Conclusion

Our paper considers the issue of anticipation effects in the Difference-in-Differences model and provides a method for identifying ATT in the presence of anticipation effects. This method can be applied in scenarios where researchers lack information regarding the timing of anticipation or only have limited data on pre-treatment periods potentially contaminated

by anticipation behavior.

We generalize the framework for partial identification in [Manski and Pepper \(2018\)](#) and develop an identification result that is robust to the violation of the non-anticipation effects. In our framework, the degree of violation depends on the choice of the bounding parameter c . We recommend that researchers use economics theory and intuition, combined with specific empirical settings, to calibrate c . We also suggest reporting robust confidence sets for the causal effect of interest under a variety of possible values of c to check the sensitivity of the results. Moreover, our framework can be adapted to accommodate multiple treatment periods, which is often encountered in empirical settings. Through empirical examples, we illustrate the practical application of our approach, providing more reliable estimates.

Overall, our methodology provides a more nuanced perspective on treatment effects in the presence of anticipation, thereby enhancing the robustness and credibility of empirical studies using DID models.

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A Proof

A.1 Proof of Lemma 1

Proof. On the one hand, we have

$$\begin{aligned}
\theta_0^{OLS} &= \mathbb{E}[Y_0|D = 1] - \mathbb{E}[Y_0|D = 0], \\
&= \mathbb{E}[Y_0(1)|D = 1] - \mathbb{E}[Y_0(0)|D = 0], \\
&= \mathbb{E}[Y_0(1)|D = 1] - \mathbb{E}[Y_0(0)|D = 1] + \mathbb{E}[Y_0(0)|D = 1] - \mathbb{E}[Y_0(0)|D = 0], \\
&= \mathbb{E}[Y_0(1) - Y_0(0)|D = 1] + \mathbb{E}[Y_0(0)|D = 1] - \mathbb{E}[Y_0(0)|D = 0], \\
&= AE + SB_0.
\end{aligned}$$

On the other hand, we have

$$\begin{aligned}
\theta_1^{OLS} &= \mathbb{E}[Y_1|D = 1] - \mathbb{E}[Y_1|D = 0], \\
&= \mathbb{E}[Y_1(1)|D = 1] - \mathbb{E}[Y_1(0)|D = 0], \\
&= \mathbb{E}[Y_1(1)|D = 1] - \mathbb{E}[Y_1(0)|D = 1] + \mathbb{E}[Y_1(0)|D = 1] - \mathbb{E}[Y_1(0)|D = 0], \\
&= \mathbb{E}[Y_1(1) - Y_1(0)|D = 1] + \mathbb{E}[Y_1(0)|D = 1] - \mathbb{E}[Y_1(0)|D = 0], \\
&= ATT + SB_1.
\end{aligned}$$

Therefore, under Assumption 1, we have $\theta^{DID} \equiv \theta_1^{OLS} - \theta_0^{OLS} = ATT - AE$. □

A.2 Proof of Proposition 1

Proof. Under Assumptions 1 and 2, we successively have

$$\begin{aligned}
-c|ATT| &\leq AE \leq c|ATT|, \\
-c|ATT| &\leq -AE \leq c|ATT|, \\
ATT - c|ATT| &\leq ATT - AE \leq ATT + c|ATT|, \\
ATT - c|ATT| &\leq \theta^{DID} \leq ATT + c|ATT|,
\end{aligned}$$

where the first inequality holds from Assumption 2, and the last holds from Lemma 1. Then, we consider two cases.

Case 1: $ATT \geq 0$

Then, $(1 - c)ATT \leq \theta^{DID} \leq (1 + c)ATT$, which implies $\frac{\theta^{DID}}{1+c} \leq ATT \leq \frac{\theta^{DID}}{1-c}$.

Case 2: $ATT < 0$

Then $(1+c)ATT \leq \theta^{DID} \leq (1-c)ATT$, which implies $\frac{\theta^{DID}}{1-c} \leq ATT \leq \frac{\theta^{DID}}{1+c}$.

Therefore, combining two cases yields

$$\min \left\{ \frac{\theta^{DID}}{1-c}, \frac{\theta^{DID}}{1+c} \right\} \leq ATT \leq \max \left\{ \frac{\theta^{DID}}{1-c}, \frac{\theta^{DID}}{1+c} \right\}.$$

□

A.3 Proof of Proposition 2

Proof. Since individuals only receive treatment in period 1,

$$\begin{aligned} AE_{t_0} &= \mathbb{E} [Y_{t_0}(\dots, D_0 = 0, D_1 = 1, D_2 = 0, \dots) - Y_{t_0}(\mathbf{0}) \mid (\dots, D_0, D_1, D_2, \dots) = (\dots, 0, 1, 0, \dots)] \\ &= \mathbb{E} [\beta^{|t_0|+1} \tau_1 \mid (\dots, D_0, D_1, D_2, \dots) = (\dots, 0, 1, 0, \dots)] \\ &= \mathbb{E} [\beta^{|t_0|+1} \mid (\dots, D_0, D_1, D_2, \dots) = (\dots, 0, 1, 0, \dots)] \cdot ATT \\ &\quad - Cov(\beta^{|t_0|+1}, \tau_1 \mid (\dots, D_0, D_1, D_2, \dots) = (\dots, 0, 1, 0, \dots)) \end{aligned}$$

By triangle inequality,

$$\begin{aligned} |AE_{t_0}| &\leq \mathbb{E} [\beta^{|t_0|+1} \mid (\dots, D_0, D_1, D_2, \dots) = (\dots, 0, 1, 0, \dots)] \cdot |ATT| \\ &\quad + |Cov(\beta^{|t_0|+1}, \tau_1 \mid (\dots, D_0, D_1, D_2, \dots) = (\dots, 0, 1, 0, \dots))| \end{aligned}$$

If β is non-random, then $|AE_{t_0}| \leq \beta^{|t_0|+1} \cdot |ATT|$ and we can take $c = \beta$.

If β is random, $\beta \perp\!\!\!\perp (Y_1(0, \dots, 0, D_1 = 1, 0, \dots), Y_t(\mathbf{0})) \mid \mathbf{D}$ and β is bounded away from 1, then $Cov(\beta^{|t_0|+1}, \tau_1 \mid (\dots, D_0, D_1, D_2, \dots) = (\dots, 0, 1, 0, \dots)) = 0$ and there exists a constant $\bar{\beta} \in (0, 1)$ s.t. $\mathbb{E} [\beta^{|t_0|+1} \mid (\dots, D_0, D_1, D_2, \dots) = (\dots, 0, 1, 0, \dots)] \leq \bar{\beta}^{|t_0|+1}$. Therefore, $|AE_{t_0}| \leq \bar{\beta}^{|t_0|+1} \cdot |ATT|$. □

A.4 Proof of Proposition 3

Proof. We have $\theta_{t_0}^{OLS} = AE_{t_0} + SB_{t_0}$. Then, under Assumption 3, we can write $\theta_{t_0}^{DID} = ATT - AE_{t_0}$.

Under Assumptions 3 and 4, we successively have

$$\begin{aligned}
-c^{|t_0|+1}|ATT| &\leq AE_{t_0} \leq c^{|t_0|+1}|ATT|, \\
-c^{|t_0|+1}|ATT| &\leq -AE_{t_0} \leq c^{|t_0|+1}|ATT|, \\
ATT - c^{|t_0|+1}|ATT| &\leq ATT - AE_{t_0} \leq ATT + c^{|t_0|+1}|ATT|, \\
ATT - c^{|t_0|+1}|ATT| &\leq \theta_{t_0}^{DID} \leq ATT + c^{|t_0|+1}|ATT|,
\end{aligned}$$

where the first inequality holds from Assumption 4, and the last holds from Lemma 1.

Case 1: $ATT \geq 0$ Then, $(1 - c^{|t_0|+1})ATT \leq \theta_{t_0}^{DID} \leq (1 + c^{|t_0|+1})ATT$ for all $t_0 \in T_0$, which implies $\frac{\theta_{t_0}^{DID}}{1+c^{|t_0|+1}} \leq ATT \leq \frac{\theta_{t_0}^{DID}}{1-c^{|t_0|+1}}$ for all $t_0 \in T_0$. Therefore,

$$\max_{t_0 \in T_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\} \leq ATT \leq \min_{t_0 \in T_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\}.$$

Case 2: $ATT < 0$

Then $(1 + c^{|t_0|+1})ATT \leq \theta_{t_0}^{DID} \leq (1 - c^{|t_0|+1})ATT$, which implies $\frac{\theta_{t_0}^{DID}}{1-c^{|t_0|+1}} \leq ATT \leq \frac{\theta_{t_0}^{DID}}{1+c^{|t_0|+1}}$ for all $t_0 \in \mathcal{T}$. Therefore,

$$\max_{t_0 \in T_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\} \leq ATT \leq \min_{t_0 \in T_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\}.$$

Hence, combining two cases yields

$$\begin{aligned}
ATT \in & \left[\max_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\}, \min_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\} \right] \cup \\
& \left[\max_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right\}, \min_{t_0 \in \mathcal{T}_0} \left\{ \frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right\} \right]
\end{aligned}$$

□

A.5 Proof of Proposition 4

Proof. We have $SB_{t_0} = \theta_{t_0}^{OLS} - AE_{t_0}$. Then, under Assumption 4, we can write:

$$\theta_{t_0}^{OLS} - c^{|t_0|+1} |ATT| \leq SB_{t_0} \leq \theta_{t_0}^{OLS} + c^{|t_0|+1} |ATT|$$

Therefore, using Assumption 5, we have

$$\min_{t_0} \{ \theta_{t_0}^{OLS} - c^{|t_0|+1} |ATT| \} \leq \min_{t_0} SB_{t_0} \leq SB_1 \leq \max_{t_0} SB_{t_0} \leq \max_{t_0} \{ \theta_{t_0}^{OLS} + c^{|t_0|+1} |ATT| \}$$

By adding ATT to each side of the equation, we have

$$\min_{t_0} \{ \theta_{t_0}^{OLS} - c^{|t_0|+1} \mid ATT \mid \} + ATT \leq SB_1 + ATT \leq \max_{t_0} \{ \theta_{t_0}^{OLS} + c^{|t_0|+1} \mid ATT \mid \} + ATT$$

Since $ATT + SB_1 = \theta_1^{OLS}$, we obtain the following identification region for the ATT :

$$\min_{t_0} \{ \theta_{t_0}^{OLS} - c^{|t_0|+1} \mid ATT \mid \} + ATT \leq \theta_1^{OLS} \leq \max_{t_0} \{ \theta_{t_0}^{OLS} + c^{|t_0|+1} \mid ATT \mid \} + ATT$$

which is equivalent to the conditions below:

$$\max_{t_0} \{ \theta_{t_0}^{DID} + c^{|t_0|+1} \mid ATT \mid \} - ATT \geq 0 \quad (\text{A.1})$$

$$\min_{t_0} \{ \theta_{t_0}^{DID} - c^{|t_0|+1} \mid ATT \mid \} - ATT \leq 0 \quad (\text{A.2})$$

If $ATT \geq 0$, Equation A.1 is equivalent to

$$\begin{aligned} & \max_{t_0 \in \mathcal{T}_0} \left\{ \left(\frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} - ATT \right) (1 - c^{|t_0|+1}) \right\} \geq 0 \\ \iff & \left(\frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} - ATT \right) (1 - c^{|t_0|+1}) \geq 0 \text{ for some } t_0 \\ \iff & \left(\frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} - ATT \right) \geq 0 \text{ for some } t_0 \text{ since } 1 - c^{|t_0|+1} \geq 0 \\ \iff & \max_{t_0 \in \mathcal{T}_0} \left(\frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} - ATT \right) \geq 0 \end{aligned}$$

Therefore, Equation A.1 is equivalent to $\max_{t_0 \in \mathcal{T}_0} \left(\frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} - ATT \right) \geq 0$. Similarly, Equation

A.2 is equivalent to $\min_{t_0 \in \mathcal{T}_0} \left(\frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} - ATT \right) \geq 0$.

Hence, if $ATT \geq 0$, we have

$$\min_{t_0 \in \mathcal{T}_0} \left(\frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right) \leq ATT \leq \max_{t_0 \in \mathcal{T}_0} \left(\frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right) \quad (\text{A.3})$$

Using a similar reasoning as above, we can show that if $ATT < 0$, we have

$$\min_{t_0 \in \mathcal{T}_0} \left(\frac{\theta_{t_0}^{DID}}{1 - c^{|t_0|+1}} \right) \leq ATT \leq \max_{t_0 \in \mathcal{T}_0} \left(\frac{\theta_{t_0}^{DID}}{1 + c^{|t_0|+1}} \right) \quad (\text{A.4})$$

The result follows from A.3 and A.4.

□

A.6 Proof of Proposition 5

Proof. Assumption 7 implies successively

$$\begin{aligned}\mathbb{E}[Y_0(0, d_1, \dots, d_T) - Y_0(0, \dots, 0) | (D_0, D_1, \dots, D_T) = (0, d_1, \dots, d_T)] &= 0, \\ \mathbb{E}[Y_0 - Y_0(0, \dots, 0) | (D_0, D_1, \dots, D_T) = (0, d_1, \dots, d_T)] &= 0, \\ \mathbb{E}[Y_0 | (D_0, D_1, \dots, D_T) = (0, d_1, \dots, d_T)] &= \mathbb{E}[Y_0(0, \dots, 0) | (D_0, D_1, \dots, D_T) = (0, d_1, \dots, d_T)].\end{aligned}$$

Combining Assumption 6 with Assumption 7 implies successively

$$\begin{aligned}\mathbb{E}[Y_t(0, \dots, 0) - Y_0 | (D_0, D_1, \dots, D_T) = (0, d_1, \dots, d_T)], \\ &= \mathbb{E}[Y_t(0, \dots, 0) - Y_0 | (D_0, D_1, \dots, D_T) = (0, \dots, 0)], \\ \mathbb{E}[Y_t(0, \dots, 0) - Y_0 | (D_0, D_1, \dots, D_T) = (0, d_1, \dots, d_T)], \\ &= \mathbb{E}[Y_t - Y_0 | (D_0, D_1, \dots, D_T) = (0, \dots, 0)], \\ \mathbb{E}[Y_t(0, \dots, 0, d_{t+1}, \dots, d_T) - Y_0 | (D_0, D_1, \dots, D_T) = (0, \dots, 0, d_{t+1}, \dots, d_T)], \\ &= \mathbb{E}[Y_t - Y_0 | (D_0, D_1, \dots, D_T) = (0, \dots, 0)], \\ \mathbb{E}[Y_t - Y_0 | (D_0, D_1, \dots, D_T) = (0, \dots, 0, d_{t+1}, \dots, d_T)], \\ &= \mathbb{E}[Y_t - Y_0 | (D_0, D_1, \dots, D_T) = (0, \dots, 0)],\end{aligned}$$

where the first equality holds from the implication of Assumption 7, the second holds from Assumption 6, the third holds by setting $(0, d_1, \dots, d_T) = (0, \dots, 0, d_{t+1}, \dots, d_T)$ and using Assumption 7, and the last holds from the definition of the potential outcome. □

A.7 Proof of Lemma 2

Proof. Given any t and s , by Assumption 6, we have

$$\begin{aligned}&\mathbb{E}[Y_t(0, \dots, 0) - Y_0(0, \dots, 0) | (D_0, D_1, \dots, D_T) = (0, \dots, d_s = 1, 1, \dots, 1)] \\ &= \mathbb{E}[Y_t(0, \dots, 0) - Y_0(0, \dots, 0) | (D_0, D_1, \dots, D_T) = (0, \dots, 0)]\end{aligned}$$

Rearranging terms gives

$$\begin{aligned}&\mathbb{E}[Y_t(0, \dots, 0) | (D_0, D_1, \dots, D_T) = (0, \dots, d_s = 1, 1, \dots, 1)] - \mathbb{E}[Y_t(0, \dots, 0) | (D_0, D_1, \dots, D_T) = (0, \dots, 0)] \\ &= \mathbb{E}[Y_0(0, \dots, 0) | (D_0, D_1, \dots, D_T) = (0, \dots, d_s = 1, 1, \dots, 1)] - \mathbb{E}[Y_0(0, \dots, 0) | (D_0, D_1, \dots, D_T) = (0, \dots, 0)]\end{aligned}$$

By definition, this implies $SB_t^s = SB_0^s$. □

A.8 Proof of Proposition 6

Proof. For each group s , we can use the information from period 0 to period s to bound both ATT_t^s and AE_{t_0} for $t_0 < s$, then utilize these bounds on AE_{t_0} for all t_0 to partially identify ATT_t^s . Again, we consider two cases.

Case 1: $ATT_s^s \geq 0$

Given $t_0 < s$, we have $-c^{s-t_0} ATT_s^s \leq AE_{t_0}^s \leq c^{s-t_0} ATT_s^s$.

Because of Assumption 6, selection bias is constant over time by Lemma 2. By Lemma 1, $AE_{t_0}^s = ATT_t^s - \theta_{t,t_0}^{DID,s}$ for all t and t_0 . Then, it follows that

$$-c^{s-t_0} ATT_s^s \leq ATT_s^s - \theta_{s,t_0}^{DID,s} \leq c^{s-t_0} ATT_s^s$$

then

$$\frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \leq ATT_s^s \leq \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}}$$

Thus, we have

$$\max_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \right\} \leq ATT_s^s \leq \min_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}} \right\}$$

By Lemma 1, we can also get the bound for $AE_{t_0}^s$,

$$\max_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \right\} - \theta_{s,t_0}^{DID,s} \leq AE_{t_0}^s \leq \min_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}} \right\} - \theta_{s,t_0}^{DID,s}$$

Given t_0 ,

$$ATT_t^s = AE_{t_0}^s + \theta_{t,t_0}^{DID,s} \in \left[\theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \max_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \right\}, \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \min_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}} \right\} \right]$$

Thus, we have

$$ATT_t^s \in \left[\max_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \max_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \right\} \right\}, \min_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \min_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}} \right\} \right\} \right]$$

Case 2: $ATT_s^s < 0$

Using analogous arguments, we can get

$$ATT_t^s \in \left[\max_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \max_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}} \right\} \right\}, \min_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \min_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \right\} \right\} \right]$$

Hence, combining two cases yields

$$ATT_t^s \in \left[\max_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \max_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \right\} \right\}, \min_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \min_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}} \right\} \right\} \right] \cup \left[\max_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \max_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 - c^{s-t_0}} \right\} \right\}, \min_{t_0} \left\{ \theta_{t,t_0}^{DID,s} - \theta_{s,t_0}^{DID,s} + \min_{t_0} \left\{ \frac{\theta_{s,t_0}^{DID,s}}{1 + c^{s-t_0}} \right\} \right\} \right]$$

□

B Additional Results

B.1 Simulation for verifying the coverage of inference procedure

We conduct Monte Carlo simulations to assess the coverage of the confidence band of ATT for different values of c . The DGP is following Example 2. For each specific c , the number of replications is 2000, with a sample size of 2000, $T_0 = 6$, $\delta = 0.5$. The result in Table B.1 shows that without doing inference, the coverage is notably low when c is around $\delta = 0.5$. By applying Nevo and Rosen’s procedure, we can improve the coverage and achieve the exact coverage when $c = \delta = 0.5$. We have also tried other DGP, and this algorithm can consistently give us a stable coverage level of around 95%, suggesting that this inferential procedure is valid within this setting.

Table B.1: Coverage of confidence band for different values of c

Values of c	0.40	0.45	0.5	0.55	0.60	0.65	0.70	0.75	0.80
Coverage without using inferential procedure	0%	0%	0%	0.4%	3.3%	31.9%	76.9%	99.2%	100%
Coverage with using Nevo’s procedure	35.1%	72.4%	95.5%	99.3%	99.9%	99.9%	100%	100%	100%

B.2 Test Non-AE and PT in Staggered Adoption Design

Example 4. (Staggered adoption design) Consider the following DGP: $\forall t = 0, \dots, T$

$$\begin{cases} Y_t = V + (\sum_{s=1}^T \theta_s D_s) \mathbb{1}\{t > 0\} + \varepsilon_t \\ D_t = \mathbb{1}\{V \geq 2 - \frac{t}{T}\} \end{cases}$$

where $U \perp (\{\varepsilon_t, \theta_t\}_{t=1}^T)$, $\theta_t \sim \mathcal{N}(\beta, 1)$, $\mathcal{T}_0 = \{0\}$, $\varepsilon_t \sim \mathcal{N}(t^2, 1)$, and $V \sim \mathcal{U}(0, 2)$.

Under the provided DGP, θ_t governs the anticipation effects and treatment effects. Let us consider the case where $T = 4$ and the number of replications is 5000.

(1) If $\beta \neq 0$ (i.e., $\mathbb{E}(\theta_t) \neq 0$), Assumption 6 holds but Assumption 7 does not hold. Table B.2 and Figure B.1 display the power of the Wald-test under varying values of β and sample sizes N . For a fixed sample size, as the mean of θ gets farther from 0, the power becomes larger and larger. For a fixed value of β , as the sample size increases, the power increases and converges to 1, so the Wald-test is consistent.

(2) If $\beta = 0$ (i.e., $\mathbb{E}(\theta_t) = 0$), both Assumption 6 and Assumption 7 hold. As shown in Table B.3, provided the mean of θ remains at 0, the size of the Wald-test can be controlled at around 5% when the sample size is large enough.

Additionally, we verify the power of the F-test as well, and the pattern closely resembles the Wald-test. The F-test can effectively control the size staying below 5% even for a small sample. But for each value of β and sample size N , the power of the Wald-test is greater than that of the F-test. Hence, both tests are valid in terms of size and power.

Table B.2: Power of Wald test

Values of β (Mean of θ_t)	Sample size N				
	100	200	500	1000	5000
0.01	8.80%	5.34%	3.58%	4.14%	4.46%
0.02	7.90%	5.30%	3.76%	3.64%	5.02%
0.05	8.06%	5.52%	5.60%	6.54%	26.00%
0.1	9.92%	8.60%	11.54%	21.46%	88.50%
0.2	17.00%	20.00%	44.02%	79.16%	100%
0.3	26.70%	41.96%	84.26%	99.04%	100%
0.5	63.82%	89.42%	99.98%	100%	100%

Notes: This table presents the power of the Wald test through simulation. Since the mean of $\theta_t \neq 0$, the extended no-anticipation effect is violated. Simulation shows that the test has a good power when the sample size is sufficiently large.

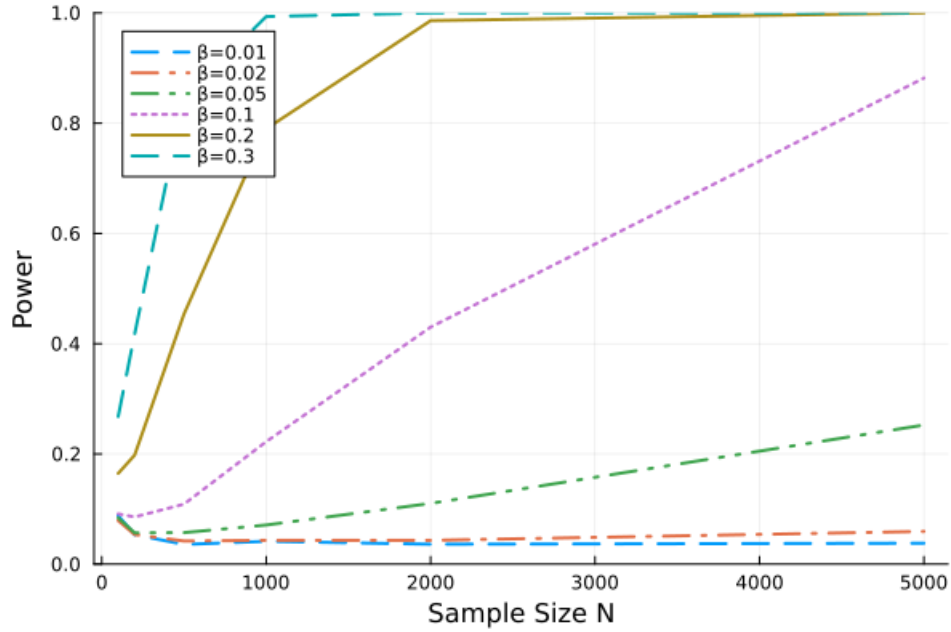


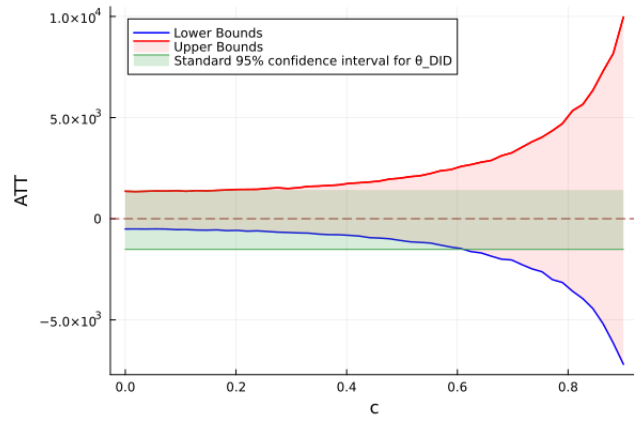
Figure B.1: Power curves of Wald test

Table B.3: Size of Wald test

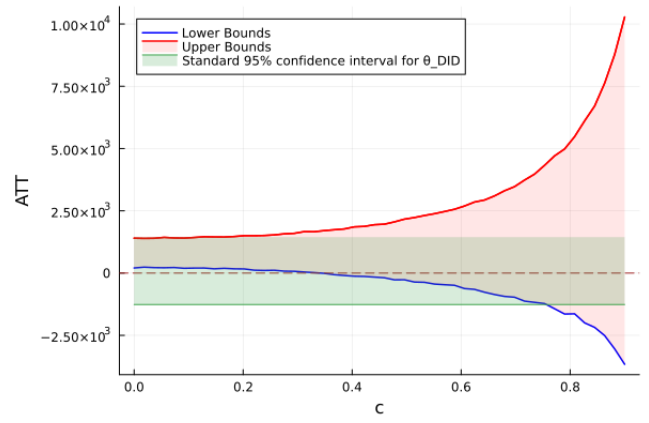
Distribution of θ_t	Sample size N				
	100	200	500	1000	5000
$\mathcal{N}(0, \frac{1}{2})$	8.90%	5.22%	3.40%	3.70%	3.82%
$\mathcal{N}(0, 1)$	8.76%	5.28%	3.40%	4.02%	4.10%
$\mathcal{N}(0, 2)$	9.02%	5.96%	4.44%	4.18%	4.98%
$\mathcal{N}(0, 4)$	10.82%	7.14%	6.04%	5.28%	6.18%
$\mathcal{U}(-\frac{1}{2}, \frac{1}{2})$	9.62%	6.44%	4.90%	4.56%	3.92%
$\mathcal{U}(-1, 1)$	8.68%	5.44%	3.74%	3.94%	3.56%
$\mathcal{U}(-2, 2)$	8.78%	5.48%	4.22%	3.86%	3.42%

B.3 Additional Empirical Result

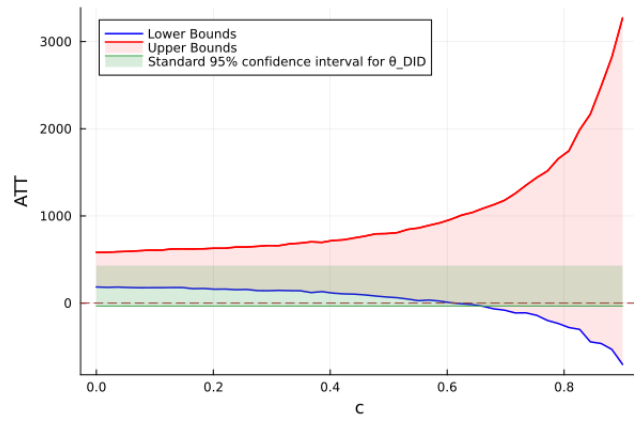
Figure B.2 presents the bounds on ATT for other outcomes categories (5) (6) (7) as reported in Kresch (2020). Plot (a) shows that the bounds contain 0 for all the values of c , indicating ATT on investment in water is not statistically significant. The finding aligns with the standard DID estimate and the TWFE estimate provided by Kresch (2020). Kresch (2020) argues that the effect on investment in water is likely attenuated by the introduction of PAC, an infrastructure development program launched in 2007, which increased investment in the water network for state-run municipalities as well. From (b) and (c) in Figure B.2, it can be observed that the ATT on investment in sewer and other investments is significantly positive for a small degree of anticipation. However, if there are large anticipation effects, the implementation effects may not be significant. In contrast, the TWFE yields significantly positive effects for both outcomes, and the standard DID gives insignificant results.



(a) Investment in Water



(b) Investment in Sewer



(c) Other Investments

Figure B.2: Bounds on ATT for categories (5) (6) (7)